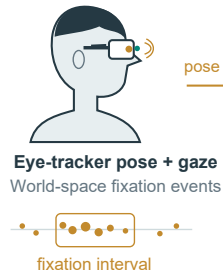


(a) Gaze + speech

Synchronized user input



Speech with word times



"Bring me that ball."

LLM: actions + slots
object = "that ball"

(d) Temporal grounding

Actions + referring expressions

(b) 3DGS-mediated instance disambiguation

Offline Multiview RGB $\rightarrow$ metric 3DGS $\rightarrow$ persistent IDs

Render at wearer pose



Depth + opacity

View-dependent surfaces

Uncertain gaze

Query a local angular patch

1 Visible surface evidence



Weighted surface mass q_k

2 Size-aware capture



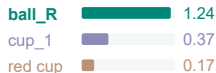
Expected capture C_k

$$\text{capture}_k = q_k / C_k$$

Normalize by target angular size

3 Rank + binding gate

Illustrative map query



ball_R or no binding

Persistent metric 3DGS map

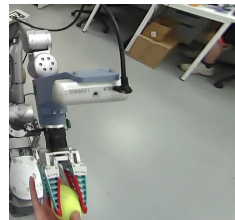


(c) Robot execution

Shared map coordinates



Navigate & re-observe



Grasp & deliver

Executable commands

action + ID + world hint

Dispatch when grounded